

Solution Code - Module 3-Core Java - StreamAPI

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q28_StreamAPI.java

```
1 | import java.util.Arrays;
2 | import java.util.List;
3 | import java.util.stream.Collectors;
4 |
5 | public class Q28_StreamAPI {
6 |     public static void main(String[] args) {
7 |         // Create a List of integers
8 |         List<Integer> numbers = Arrays.asList(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 22, 33, 44);
9 |
10 |        System.out.println("Original numbers list: " + numbers);
11 |
12 |        // Filter even numbers and collect the result
13 |        List<Integer> evenNumbers = numbers.stream()
14 |            .filter(n -> n % 2 == 0)
15 |            .collect(Collectors.toList());
16 |
17 |        System.out.println("Filtered even numbers list: " + evenNumbers);
18 |
19 |        // Additional stream processing: printing them individually using forEach
20 |        System.out.print("Even numbers formatted: ");
21 |        numbers.stream()
22 |            .filter(n -> n % 2 == 0)
23 |            .forEach(n -> System.out.print "[" + n + " "));
24 |        System.out.println();
25 |    }
26 | }
```